

Fundamentals of BioEngineering (FunBio)

ChE 338

F2025

Class

MWF 12 – 12:50 pm
CPE 2.214

#16850

Instructor

JENNIFER MAYNARD
Office hours

maynard@che.utexas.edu
Tues 3-4 pm in CPE 5.466

*cc TA on all emails please

TA

Office hours
CHARLIE JOHNSON

Thurs 3-5 pm, CPE 5.462
cdjohnson98@utmail.utexas.edu

Contact for questions related to homework, submission logistics (uploading submissions on time, online circumstances), office hours, project scope and execution

Tutors & Graders

Tutor: Esdras Lopez

esdras.e.lopez@gmail.com

6 hr/ week

Grader: Emily Peterson

emilyp.home@gmail.com

Textbook

- The HW, quiz and super-quiz materials will be based primarily on lectures and in-class examples. To supplement this, readings from the *Wittrup textbook below* will be assigned and ones from other sources will be provided on Canvas.
- *Quantitative Fundamentals of Molecular and Cellular Bioengineering* by Wittrup, Tidor, Hackel and Sarkar (\$29 to rent; \$67 to buy on Amazon). **we will use this for about half the class*
- Optional: [Fundamentals of Biochemistry](#) by Jakubowski and Flatt, 1st edition, LibreTexts (free)

Overview

This course is designed for students with an engineering background to learn the biological fundamentals of biotechnology using chemical engineering analyses. The course emphasizes conceptual appreciation of the molecular interplays which are the basis of “chemical processes” in living systems. The objective of the course is to provide students with a comprehensive and concise overview of molecular biology, cellular and protein science relevant to chemical engineers while learning to apply quantitative engineering principles, including thermodynamics, kinetics and stoichiometry, to the analysis of biological processes.

Course goals

To provide chemical engineering students with a basic understanding of fundamental biological principles and a working vocabulary that would enable them to expand their knowledge base during their academic and professional careers. To convey to students an appreciation of the wide variety of engineering applications that are related to the fields of biochemistry, cell biology, genetics, general microbiology, and environmental microbiology.

Objectives

- Apply material balances to biological processes such as transcription, translation, cellular growth and spread of infection.
- Apply mass action chemical kinetics to nucleic acid annealing, protein aggregation, ligand-protein binding and enzyme functions.
- Apply thermodynamics to nucleic acid annealing, protein-ligand binding and protein aggregation processes.
- Understand structural basis of molecular recognition in biological systems
- An ability to fit data to simple kinetic and thermodynamic models and use these to predict future behavior and define design objectives.
- An appreciation for the history of biochemical engineering, including the contributions of diverse scientists and key scientific breakthroughs that have had dramatic impacts on humanity.
- An appreciation of the professional opportunities to apply chemical engineering skills to biotechnologies and an appreciation for the ethical and policy implications of some technologies.

Knowledge, abilities and skills students should have

Prerequisites: completion of ChE 317. Material balances, thermodynamics and kinetics will be employed, but will also be reviewed before first used.

- You should have a working knowledge of algebra and the ability to solve simple ordinary differential equations, *i.e.*, M408C/D at UT or the equivalent with a grade of at least C.
- ChE 317: material balances on 1-3 units at steady-state and un-steady-state.
- Chem 210 and OChem: basic structure/ function reactivities to understand amino acid and nucleic acid chemistry, protein structure and function and metabolism.
- ChE 322 Thermo: determine free energies of reaction (ΔG) and convert these to enthalpies (ΔH) and entropy (ΔS) or equilibrium K_d or K_a values; determine the extent of reaction at equilibrium.

Chemical engineering program outcomes achieved

- ABET student outcome 1: An ability to identify, formulate, and solve complex engineering problems by applying principles of engineering, science, and mathematics.

Course procedure

- Canvas will be used to post homework assignments, assignment grades, reading material and any announcements. Please check Canvas at least weekly.
- Class time is where concepts will be explained, practice problems worked and information about HW, quizzes and projects shared. The general format will be an advance in biochemical engineering, the biology background needed to understand this and the associated engineering analyses to design/optimize with an example problem and solution. Insta-Poll questions will be used to gauge attendance.
- Participation: Insta-poll will be used approximately once per class period, with two drops.
- Office hours are to get more information or a different perspective on material and ask questions about material. We will provide additional office hours in anticipation of HW and super-quizzes.
- Gradescope will be used to download, submit and anonymously grade (and re-grade) most assignments.
- Quizzes: weekly short quizzes to cover the prior week's reading, lecture and HW assignment (if one was due). Your lowest quiz score will be dropped; if you need to miss a quiz for any reason, this will be your dropped quiz. These will be held on Fridays in class.
- Homework: will be due roughly every other week throughout the semester.
- Super-quizzes: The three scheduled super-quizzes will include problems similar to those worked in class and on homework and will require knowledge of the relevant biology. There are no make-up superquizzes. If you have an excused/ essential conflict, you may elect take a cumulative final superquiz during the regularly scheduled final period in lieu of the superquiz you missed.
- Late work policy: Late times for HWs will be noted in Gradescope and will incur a 10% penalty. Any timed assessment that is submitted ≥ 5 minutes late will incur a penalty proportional to the late-ness

(e.g., for a 30 min quiz is submitted 30 min late, only 50% of the time used was allowed and thus only 50% of the points earned will be provided to the student). We are unable to accept any assignment submitted after the solutions have been posted.

- **Regrade policy:** Regrades requests are due via Gradescope 1 week after the graded item is returned. Regrades will be performed by the person who originally graded it using the same rubric to ensure consistent grading across all students. This may trigger regrading of the entire submission which could result in a lower overall score.
- **Project** (in lieu of final exam): Students (2-4 per group) will prepare a presentation about a “feat of bioengineering” – a technology or therapeutic related to the class and will explain the basic biology and engineering approaches to either understand or assist development of the technology. The presentation (~12 min) will include an overview of the project goals, approach and key data, as well as comments on the project’s significance, weaknesses and likelihood of success. Topics may be selected from a suggested list or a student-selected topic. To manage the class size, these will be recorded and students will watch assigned projects and answer a related quiz question.
- **Extra credit** will be provided for introducing a “Super Scientist” in the bio-engineering space to the class via 1-2 slides by 1-2 students in ~5 min. Opportunities to do this will be noted in advance in class; a list of suggested topics will be provided. This will add up to 0.3% to your final weighted average. You can receive extra credit once and this is the only opportunity for extra credit.

Grading

Final grades will be based on the following weighted percentages and will include +/- grades according to the standard cut-offs below. Grades may be curved (grade cut-offs lowered) but they will not be raised.

Quizzes (weekly on Fri)	15% (cover lecture content & short problems; 1 drop)
Homework (~bi-weekly)	15% (~6-8 sets total)
Insta-poll (in class)	10% (periodically, drop 2)
Super quizzes (3)	45% (15% each)
<u>Project presentation</u>	<u>15% (in lieu of final; includes project quizzes)</u>
Total	100%

< 59	60-62	63-66	67-69	70-72	73-76	77-79	80-82	83-86	87-89	90-92	93-100
F	D-	D	D+	C-	C	C+	B-	B	B+	A-	A

Key due dates

- Superquiz 1: Fri, Sept 26, 2025
- Superquiz 2: Fri, Oct 24, 2025
- Project due: Mon Nov 24 (soft due date with a hard due date Mon Dec 8)
- Fall break: Nov 24-28
- Superquiz 3: Fri Dec 5, 2025
- Final exam: Th Dec 11, 1:00-3:00 pm (projects watch assignments must be completed by this time)
- Grades due: Mon Dec 15, 2025, 11:59 pm.

Typical weekly rhythm

- Class MWF 12-1 pm
- HW due alternate Wed at 10 pm on Gradescope
- Th noon: post HW solutions on Canvas ~12 pm
- Friday weekly quiz in class

Presentation plans

Planning will begin after SuperQuiz 1 is returned and are due before Fall break with watch assignments to be completed during our regularly scheduled final exam time.

Course organization

1. Overview of basic cell and molecular biology concepts (Week 1)
Review key math, chemistry and chemical engineering concepts
Biological building blocks
Cellular structure & organization
2. Cell growth and bioreactors (Weeks 2-4)
Exponential cell growth models
Monod growth models
Compartment models of cell differentiation
Bioreactors: chemostats, sterilization, washout
3. Cell metabolism & metabolic engineering (Weeks 6-9)
Building synthetic cells; cell material balances and carbon sources
Cellular energetics and electron sources
Overview of cellular metabolism
Enzyme kinetics, competitive and non-competitive inhibition (Michaelis-Menton)
Metabolic engineering concepts
Flux balance analyses for metabolic engineering
4. Gene expression and related technologies (Weeks 10-11).
Nucleic acids: structures, annealing thermodynamics and kinetics
Transcription: mRNA material balances and regulation
Translation: protein material balances and control features
5. Protein folding and function (Weeks 11-15)
Protein structure, analysis and prediction (e.g., alpha-fold)
Protein folding & thermodynamics
Protein aggregation & kinetics
Molecular recognition: binding kinetics
Drug delivery and pharmacokinetics

Official notes

Classroom policies

- Our physical and virtual classrooms are safe places for sharing opinions and thoughts in which we will all show our respect for each other by engaging without judgement, regardless of race, culture, ethnicity, religion, politics, gender or gender variance and nationality. This year more than ever is a time for us all to practice patience and kindness with ourselves and with others.
- My goal is to have you all learn and see the relevance of the course topics to your own lives and your chosen profession. If there are aspects of this course that prevent you from learning or exclude you, please let me know as soon as possible so we can jointly identify strategies that meet your needs and the requirements of the course. Resources which may be helpful are noted in the following sections.
- If you experience any specific hardships, such as illness, accident or family crisis please know that these policies may be amended. In this situation, please communicate with me, a TA or the ChE Undergraduate Advisor Brandon Fris as soon as you feel comfortable. If you do not feel comfortable discussing with us, please visit Student Emergency Services.

Academic integrity Each student in the course is expected to abide by the University of Texas Honor Code: "As a student of the University of Texas at Austin, I shall abide by the core values of the University and uphold academic integrity." Plagiarism is taken very seriously at UT. You are responsible for understanding UT's Academic Honesty and the University Honor Code which can be found at the following

web address: <https://deanofstudents.utexas.edu/conduct/standardsconduct.php>. All suspected misconduct will be formally referred to Student Conduct and Academic Integrity (SCAI) in the Office of the Dean of Students.

Acceptable use policy for AI, including ChatGPT. All assignments should be fully prepared by the student. Developing strong competencies in the skills associated with this course, from student-based brainstorming to project development, will prepare you for success in your degree program and, ultimately, a competitive career. Therefore, the use of generative AI tools to complete any aspect of assignments for this course are not permitted and will be treated as plagiarism except when this is introduced as a problem-solving strategy, e.g., the use of AlphaFold to predict protein structures. If you have questions about what constitutes a violation of this statement, please contact me.

<https://security.utexas.edu/ai-tools>; <https://ctl.utexas.edu/5-things-know-about-chatgpt>

Dropping the course An engineering student must have the Dean's approval to add or drop a class after the 4th class day of the semester. Adds or drops are not approved after the fourth class day except for good cause. "Good cause" is determined to be documented evidence of an extenuating circumstance (such as health or personal problems) that did not exist before the 4th day of class. Applications for approval to drop a course after the 4th day should be made in the Office of Student Affairs, ECJ 2.200.

If you want to drop a class after the 12th class day, you'll need to execute a Q-drop before the Q-drop deadline, which typically occurs near the middle of the semester. Under Texas law, you are only allowed six Q drops while you are in college at any public Texas institution. For more information, see: <http://www.utexas.edu/ugs/csacc/academic/adddrop/qdrop>

Religious holidays By UT Austin policy, you must notify the instructor of your pending absence at least fourteen days prior to the date of observance of a religious holy day. If you must miss a class, an examination, a work assignment, or a project in order to observe a religious holy day, you will be given an opportunity to complete the missed work within a reasonable time after the absence.")

COVID-related information

- We respect the decision made by class members to wear or not wear masks as is appropriate for each individual's personal situation.
- For the most up to date information on learning during COVID-19, including guidance about safety, masks in classrooms, and classroom guidelines, see <https://protect.utexas.edu/learning/>.

Emergency evacuations The following recommendations regarding emergency evacuation from the Office of Campus Safety and Security, 512-471-5767, <http://www.utexas.edu/safety/>.

Disability & access The university is committed to creating an accessible and inclusive learning environment consistent with university policy and federal and state law. Please let me know if you experience any barriers to learning so I can work with you to ensure you have equal opportunity to participate fully in this course. For more information, see <https://diversity.utexas.edu/disability/>.

Counseling and Mental Health Center The Counseling and Mental Health Center at <https://cmhc.utexas.edu/index.html> being, academic and life goals. To learn more about your counseling and mental health options, call CMHC at (512) 471-3515. If you are experiencing a mental health crisis, call the CMHC Crisis Line 24/7 at (512) 471-2255.

The Sanger Learning Center Did you know that more than one-third of UT undergraduate students use the Sanger Learning Center each year to improve their academic performance? All students are welcome to take advantage of the Sanger Center's workshops, private learning specialist appointments, peer academic coaching, and tutoring for more than 70 courses in 15 different subject areas. For more information, please visit <http://www.utexas.edu/ugs/slc> or call 512-471-3614 (JES A332).

BeVocal is a university-wide initiative to promote the idea that individual Longhorns have the power to prevent high-risk behavior and harm. At UT Austin all Longhorns have the power to intervene and reduce harm. To learn more about BeVocal and how you can help to build a culture of care on campus, go to: <https://wellnessnetwork.utexas.edu/BeVocal>.

If you have concerns about the safety or behavior of fellow students, TAs or Professors, call BCAL (the Behavior Concerns and COVID-19 Advice Line): 512-232-5050. Your call can be anonymous. If something doesn't feel right – it probably isn't. Trust your instincts and share your concerns.

Land acknowledgment We would like to acknowledge that we are meeting on Indigenous land. Moreover, we would like to acknowledge and pay our respects to the Carrizo & Comecrudo, Coahuiltecan, Caddo, Tonkawa, Comanche, Lipan Apache, Alabama-Coushatta, Kickapoo, Tigua Pueblo, and all the American Indian and Indigenous Peoples and communities who have been or have become a part of these lands and territories in Texas, here on Turtle Island.